

CS173: Discrete Structures

Problem Set 6

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1. Prove $(\forall a, b, x, y, n \in \mathbb{Z})((n \mid x \wedge n \mid y) \Rightarrow (n \mid ax + by))$.

Proof. Let $a, b, x, y, n \in \mathbb{Z}$ be arbitrary, and suppose $n \mid x$ and $n \mid y$.

By definition of divisibility, there exist integers $k_1, k_2 \in \mathbb{Z}$ such that $x = nk_1$ and $y = nk_2$. Then:

$$\begin{aligned} ax + by &= a(nk_1) + b(nk_2) \\ &= n(ak_1) + n(bk_2) \\ &= n(ak_1 + bk_2) \end{aligned}$$

Since $\mathbb{Z}$ is closed under addition and multiplication, $ak_1 + bk_2 \in \mathbb{Z}$.

Therefore $n \mid ax + by$ by definition of divisibility. Since a, b, x, y, n were arbitrary, we conclude $(\forall a, b, x, y, n \in \mathbb{Z})((n \mid x \wedge n \mid y) \Rightarrow (n \mid ax + by))$.

2. We say an integer $z \in \mathbb{Z}$ is **even** $:\Leftrightarrow 2 \mid z$, and we say z is **odd** $:\Leftrightarrow 2 \mid z - 1$.

- (a) Prove that no integer can be both even and odd at the same time.

Proof. Suppose for contradiction that some $z \in \mathbb{Z}$ is both even and odd.

Then $2 \mid z$ and $2 \mid z - 1$, so there exist $k_1, k_2 \in \mathbb{Z}$ such that $z = 2k_1$ and $z - 1 = 2k_2$. Substituting the first into the second:

$$\begin{aligned} 2k_1 - 1 &= 2k_2 \\ 2(k_1 - k_2) &= 1 \end{aligned}$$

Since $\mathbb{Z}$ is closed under subtraction, $k_1 - k_2 \in \mathbb{Z}$, this gives $2 \mid 1$, which contradicts the fact that $1 < 2$. Therefore no integer can be both even and odd.

- (b) Prove that $(\forall z \in \mathbb{Z})((z \text{ is even}) \Leftrightarrow (z + 1 \text{ is odd}))$.

Proof. Let $z \in \mathbb{Z}$ be arbitrary.

$(\Rightarrow)$ Suppose z is even, i.e. $2 \mid z$. We must show $z + 1$ is odd, i.e. $2 \mid (z + 1) - 1$. But $(z + 1) - 1 = z$, and $2 \mid z$ by assumption. Therefore $z + 1$ is odd.

$(\Leftarrow)$ Suppose $z + 1$ is odd, i.e. $2 \mid (z + 1) - 1$. But $(z + 1) - 1 = z$, so $2 \mid z$. Therefore z is even.

Since z was arbitrary, $(\forall z \in \mathbb{Z})((z \text{ is even}) \Leftrightarrow (z + 1 \text{ is odd}))$.

(c) Prove that $(n \text{ is even}) \vee (n \text{ is odd})$ for all $n \in \mathbb{N}$.

Proof. We proceed by weak induction on n .

Basis Step: We must show 0 is even or 0 is odd. Since $0 = 2 \cdot 0$ and $0 \in \mathbb{Z}$, we have $2 \mid 0$, so 0 is even.

Inductive Hypothesis: Let $k \in \mathbb{N}$ and assume k is even or k is odd.

Inductive Step: We must show $k + 1$ is even or $k + 1$ is odd. We consider two cases.

Case 1: Suppose k is even. By part (b), $k + 1$ is odd.

Case 2: Suppose k is odd, so $2 \mid k - 1$. Then there exists $m \in \mathbb{Z}$ such that

$$k - 1 = 2m$$

$$k = 2m + 1$$

$$k + 1 = 2m + 2$$

$$k + 1 = 2(m + 1)$$

Since $m + 1 \in \mathbb{Z}$, we have $2 \mid k + 1$, so $k + 1$ is even.

Therefore, by weak induction, $(n \text{ is even}) \vee (n \text{ is odd})$ for all $n \in \mathbb{N}$.

3. Show that $(\exists m \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq m \Rightarrow n^2 < 2^n)$.

Proof. By inspection of small cases, $m = 5$ suffices. We claim $m = 5$ witnesses the existential.

We first establish a lemma.

Lemma: $(\forall k \in \mathbb{N})(k \geq 5 \Rightarrow 2k + 1 < k^2)$.

Proof of Lemma. We proceed by weak induction on k .

Basis Step: We have $2(5) + 1 = 11 < 25 = 5^2$.

Inductive Hypothesis: Let $k \in \mathbb{N}$ with $k \geq 5$, and assume $2k + 1 < k^2$.

Inductive Step: We must show $2(k + 1) + 1 < (k + 1)^2$. Observe:

$$\begin{aligned} 2(k + 1) + 1 &= 2k + 3 \\ (k + 1)^2 &= k^2 + 2k + 1 \end{aligned}$$

So it suffices to show $2k + 3 < k^2 + 2k + 1$, i.e. $2 < k^2$. Since $k \geq 5$, we have $k^2 \geq 25 > 2$. Therefore $2(k + 1) + 1 < (k + 1)^2$. $\square$

We proceed by weak induction on n .

Basis Step: We have $5^2 = 25 < 32 = 2^5$.

Inductive Hypothesis: Let $k \in \mathbb{N}$ with $k \geq 5$, and assume $k^2 < 2^k$.

Inductive Step: We must show $(k + 1)^2 < 2^{k+1}$. Observe:

$$\begin{aligned} (k + 1)^2 &= k^2 + 2k + 1 \\ &< 2k^2 && \text{by the Lemma, since } 2k + 1 < k^2 \\ &< 2 \cdot 2^k && \text{by the inductive hypothesis} \\ &= 2^{k+1} && \text{by definition of exponentiation} \end{aligned}$$

Therefore $(k + 1)^2 < 2^{k+1}$.

By induction, $(\forall n \in \mathbb{N})(n \geq 5 \Rightarrow n^2 < 2^n)$, and so $m = 5$ witnesses $(\exists m \in \mathbb{N})(\forall n \in \mathbb{N})(n \geq m \Rightarrow n^2 < 2^n)$.

4. The factorial function $\cdot! : \mathbb{N} \rightarrow \mathbb{N}$ is the recursive function given below for all $n \in \mathbb{N}$.

$$0! := 1$$

$$(n + 1)! := (n!) \cdot (n + 1)$$

Prove that $(\forall n \in \mathbb{N})(n \geq 4 \Rightarrow 2^n < n!)$.

Proof. We will prove by weak induction on n .

Basis Step: We must show $2^4 < 4!$. We first expand $4!$ using the recursive definition.

$$\begin{aligned} 4! &= (((1) \cdot 1) \cdot 2) \cdot 3 \cdot 4 && \text{by definition of !} \\ &= 24 && \text{by arithmetic} \end{aligned}$$

Since $2^4 = 16 < 24 = 4!$, the basis step holds.

Inductive Hypothesis: Let $k \in \mathbb{N}$ with $k \geq 4$, and assume $2^k < k!$.

Inductive Step: We must show $2^{k+1} < (k + 1)!$. Observe:

$$\begin{aligned} 2^{k+1} &= 2^k \cdot 2 && \text{by definition of exponentiation} \\ &< k! \cdot 2 && \text{by the inductive hypothesis} \\ &< k! \cdot (k + 1) && \text{since } k \geq 4 \text{ implies } k + 1 > 2 \\ &= (k + 1)! && \text{by definition of !} \end{aligned}$$

Therefore $2^{k+1} < (k + 1)!$.

By induction, $(\forall n \in \mathbb{N})(n \geq 4 \Rightarrow 2^n < n!)$.

5. The Fibonacci sequence is the function $\mathcal{F} : \mathbb{N} \rightarrow \mathbb{N}$ defined below for all $n \in \mathbb{N}$.

$$\mathcal{F}(0) := 0$$

$$\mathcal{F}(1) := 1$$

$$\mathcal{F}(n+2) := \mathcal{F}(n+1) + \mathcal{F}(n)$$

Prove that $\sum_{i=1}^n \mathcal{F}(2i-1) = \mathcal{F}(2n)$ for all $n \in \mathbb{N}$.

Proof. We proceed by weak induction on n .

Basis Step: We must show $\sum_{i=1}^0 \mathcal{F}(2i-1) = \mathcal{F}(0)$. The left side is an empty sum, which equals 0 by convention. The right side is $\mathcal{F}(0) = 0$ by definition. Therefore both sides are equal.

Inductive Hypothesis: Let $k \in \mathbb{N}$ and assume $\sum_{i=1}^k \mathcal{F}(2i-1) = \mathcal{F}(2k)$.

Inductive Step: We must show $\sum_{i=1}^{k+1} \mathcal{F}(2i-1) = \mathcal{F}(2k+2)$. Observe:

$$\begin{aligned} \sum_{i=1}^{k+1} \mathcal{F}(2i-1) &= \left(\sum_{i=1}^k \mathcal{F}(2i-1) \right) + \mathcal{F}(2(k+1)-1) && \text{by definition of iterated summation} \\ &= \mathcal{F}(2k) + \mathcal{F}(2(k+1)-1) && \text{by the inductive hypothesis} \\ &= \mathcal{F}(2k) + \mathcal{F}(2k+1) && \text{by arithmetic} \\ &= \mathcal{F}(2k+2) && \text{by definition of } \mathcal{F} \end{aligned}$$

Therefore $\sum_{i=1}^{k+1} \mathcal{F}(2i-1) = \mathcal{F}(2k+2)$.

By induction, $\sum_{i=1}^n \mathcal{F}(2i-1) = \mathcal{F}(2n)$ for all $n \in \mathbb{N}$.